

NOTES

Mapping of the Radiofrequency Field

E. K. INSKO AND L. BOLINGER

Department of Radiology, MMRRCC, University of Pennsylvania, Basement, Blockley Hall, Philadelphia, Pennsylvania 19104-6021

Received February 20, 1992; revised July 29, 1992

In addition to providing an excellent method for evaluating NMR coil performance, a map of the B_1 field of a surface or volume coil would allow one to correct for many of the problems associated with nonuniform B_1 fields. First, nonuniform B_1 fields are a source of problems in both surface- and volume-coil imaging. Surface-coil image intensity drops off severely with increasing distance from the coil because of the extreme B_1 inhomogeneity. To the extent that volume coils have nonuniform B_1 fields, the images taken with volume coils will also exhibit nonuniform signal intensities. If the B_1 map of a surface or volume coil is known, the signal intensity may be corrected so that it is uniform across the image. Second, nonuniform B_1 fields make it difficult to make quantitative comparisons of signals obtained from different voxels in localized NMR spectroscopy. This is because the signal intensity is not proportional to the concentration of nuclei alone. The signal is also affected by both the strength of B_1 and any B_1 inhomogeneity present. However, if the B_1 field is known, the signal intensity of localized spectra may be normalized so that valid, quantitative signal comparisons may be made between various localized regions. This paper presents a new method for measuring the B_1 map in addition to presenting some useful results of the field mapping when applied to coil design.

The NMR probe is designed to function both as an antenna for transmission and as an antenna for reception. The amount of transverse magnetization induced in the sample by the probe transmission is dependent on sample characteristics as well as probe characteristics. The sample-dependent characteristics, concentration of nuclei and relaxation rates, determine the total amount of magnetization. The probe transmission affects only the amount of magnetization rotated into the transverse plane. In addition, the reception sensitivity of the NMR probe is directly proportional to the B_1 field that it generates. Therefore, in a gradient-echo image, the signal is proportional to the magnetization vector length, the strength of the B_1 field, and the sine of the flip angle. This signal may be divided into parts that are known and parts that are not known. Typically, nothing is known a priori about the sample. Thus, the length of magnetization

vector is unknown. The strength of the B_1 field and other coefficients of the signal strength are also unknown. Only the amount of time that the B_1 field was applied and, with a volume coil, a rough estimate of the flip angle are known.

In order to obtain the B_1 map from the NMR signal, some of the unknowns must first be eliminated. This is accomplished by taking the ratio of two different images. Taking the ratio eliminates the unknown coefficients of the signal relating to characteristics of the sample and the reception sensitivity of the NMR probe. What remains in a fully relaxed sample is the ratio of two trigonometric functions. In a $\theta_1 - \tau - 2\theta_1 - \tau$ spin-echo image and a $\theta_2 - \tau - 2\theta_2 - \tau$ spin-echo image, each using a phase-cycling scheme as given in Table 1 of Ref. (1), the following signal ratio is produced:

$$r = \sin^3 \theta_1 / \sin^3 \theta_2. \quad [1]$$

The spin echo is advantageous because it refocuses T_2^* , thereby eliminating signal variability due to inhomogeneities in the B_0 field. The phase cycling is convenient because it eliminates signals other than the signal from the spin echo. This is critical because flip angles will purposefully be chosen such that $\theta_2 = 2\theta_1$. Without phase cycling, the signal collected during the spin echo could contain a residual signal from either of the two spin-echo pulses. The $\sin^3 \theta$ flip-angle dependence of the signal (2) also makes the B_1 mapping very sensitive to small changes in B_1 because it is a relatively steep function in regions away from flip angles of 0° , 180° , 360° , etc. (see Fig. 1). Thus, this method is well suited for generating the B_1 field maps for volume coils. With the flip angles chosen such that $\theta_2 = 2\theta_1$ the signal ratio reduces to

$$r = 1/[2 \cos \theta_1]^3. \quad [2]$$

This function is easily soluble for θ_1 . Using $\theta_1 = \gamma B_1 t$, the field of the volume coil is given by

$$B_1 = (1/\gamma t) \cos^{-1}[(1/8r)^{1/3}]. \quad [3]$$

Instead of two spin-echo images, the ratio of two different-

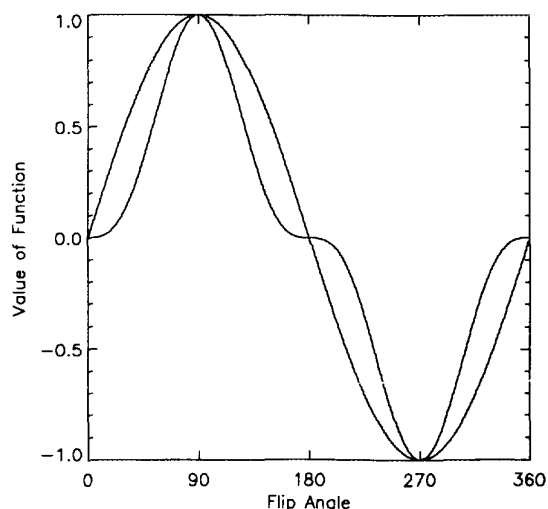


FIG. 1. The functions $\sin \theta$ and $\sin^3 \theta$ are the flip-angle dependences of the gradient-echo and the phase-cycled spin-echo sequences used to generate the B_1 maps for the surface and volume coils, respectively.

flip-angle gradient-echo images is used to map the field of a surface coil. The phase-cycled spin-echo sequence used with the volume coils would have had the advantage of refocusing T_2^* but would have also generated a $\sin^3 \theta$ flip-angle dependence of the signal. The $\sin^3 \theta$ flip-angle dependence is a disadvantage for the surface coil for exactly the same reason that it is an advantage for the volume coil. The $\sin^3 \theta$ function is rapidly varying between 20° and 160° but very flat at 0° and 180° . A gradient echo was used because it generates a simple and less rapidly varying $\sin \theta$ flip-angle dependence of the signal (see Fig. 1). This is expedient for mapping the B_1 field from surface coils, in which, unlike volume coils, the field varies rapidly as a function of space. The $\sin \theta$ flip-angle dependence of the signal therefore produces a B_1 map which is sensitive to a wide range of B_1 values for surface coils. The ratio of the signals from two different flip-angle gradient-echo images in a fully relaxed sample is given by

$$r = \sin \theta_1 / \sin \theta_2. \quad [4]$$

With the pulse lengths chosen such that $\theta_2 = 2\theta_1$ the signal ratio of the surface-coil images is given by

$$r = 1/[2 \cos \theta_1]. \quad [5]$$

Using $\theta_1 = \gamma B_1 t$, the field of the surface coil is given by

$$B_1 = (1/\gamma t) \cos^{-1}(1/2r). \quad [6]$$

A volume-coil B_1 map was generated in a double-tuned discrete cosine coil (3) and the results are shown in Fig. 2. The flip angles chosen for the two spin-echo images were

60° – 120° and 120° – 240° , both because they are symmetric around the 90° – 180° spin-echo flip-angle pair and because they lie on the steep part of the $\sin^3 \theta$ signal function, thereby ensuring that the B_1 map will be sensitive to small changes in B_1 . The images were collected in 1 cm thick 10 mM copper sulfate in water phantoms with hard, nonselective pulses at an echo time of 22 ms and a repetition time of 1 s. The results of the B_1 map indicate that the discrete cosine coil has excellent homogeneity and most of the inhomogeneity is due to increased current in the outermost wires of the coil. The increased current in the outer struts is due to the lower mutual inductance of those struts when compared to the central struts. The homogeneity of the discrete cosine coil improved dramatically when it was inductively driven at the inductor in the double-tuning trap (4). Driving the coil by inductively coupling into the main body of the coil distorted the B_1 field notably. A second volume-coil map was generated using the same method and parameters in a new biplanar coil design (5) and is shown in Fig. 3. The B_1 map of the biplanar coil exhibits the exceptional homogeneity of this simple, new design. In addition, it was discovered that the homogeneity could be maintained in a single-tuned coil by placing a loop in one of the symmetrically placed wires in the coil and driving it inductively at that point.

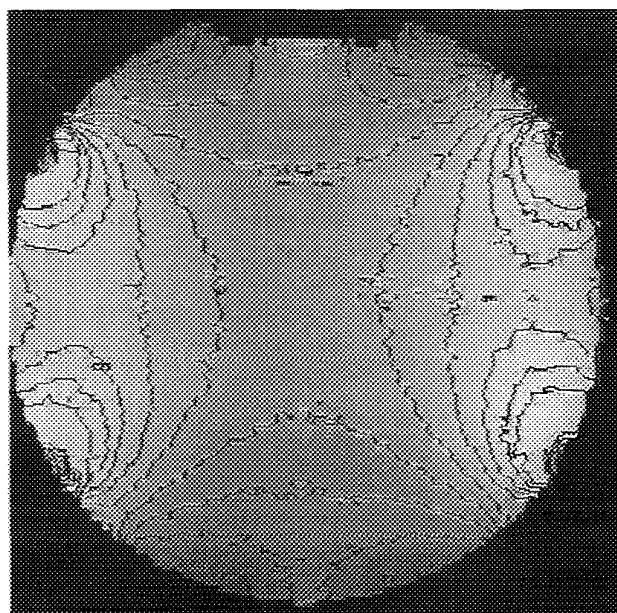


FIG. 2. The B_1 maps of a 22-strut, 13.5 cm diameter, double-tuned, discrete cosine coil, made using a 1 cm thick disk phantom filled with 10 mM copper sulfate in water. The images used to construct the map were made at an echo time of 22 ms and a repetition time of 1 s. The coil was inductively driven at the inductor in the double-tuning trap. The value of the central field is 0.21 G. The contours are separated by 0.018 G. The wires of the coil were 2 mm from the edge of the water in the phantom. Note the inhomogeneity due to the higher current on the outermost wires of the coil. The dip at the top of the map is due to the meniscus in the phantom.

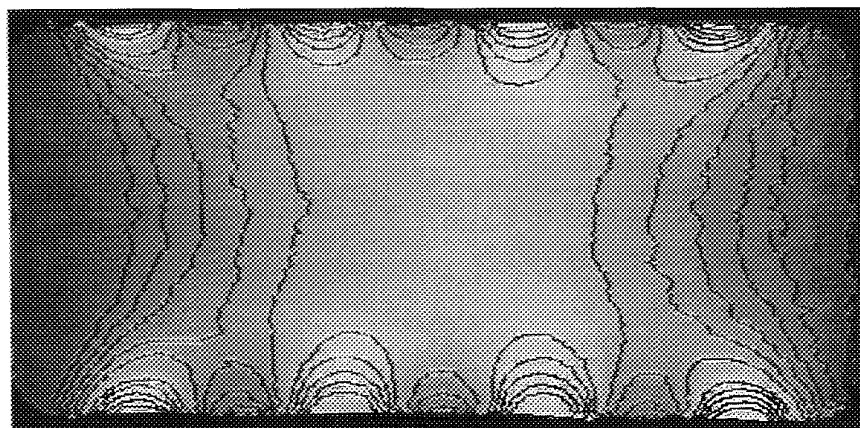


FIG. 3. The B_1 map of an eight-strut, single-tuned biplanar coil made using a 1 cm thick rectangular phantom filled with 10 mM copper sulfate in water. The images used to construct the map were made at an echo time of 22 ms and a repetition time of 1 s. The coil was inductively driven at a symmetrically placed loop. The value of the central field is 0.23 G. The contours are separated by 0.0115 G. The wires of the coil were 6 mm from the edge of the water in the phantom. Note the large homogeneous region in the center of the coil.

The B_1 map of a surface coil was generated and the results are shown in Fig. 4. The surface coil was double tuned and inductively driven at the LC trap inductor. The images were collected in 1 cm thick 10 mM copper sulfate in water phantoms with hard, nonselective pulses at an echo time of 22 ms and a repetition time of 1 s. Construction of the surface-coil map required more attention to detail since θ_1 flip angles of 90° create computational difficulties in generating the B_1 map because the signal ratio given in Eq. [5] goes to infinity. It was discovered that if the $\theta_2 = 2\theta_1$ and the θ_1 flip-angle pair of images was selected such that the θ_2 flip angle was strong enough to create just less than a 180° flip at the center of the coil, then the map could be constructed with the least difficulty.

Of course, the regions of the B_1 map close to the wires in the surface coil will always suffer from the problem of receiving large levels of B_1 and thus θ_1 flip angles near 90° , 270° , etc., creating additional divergence problems, or angles larger than 180° , creating problems with the single-valuedness of the arccosine function. These problems were eliminated to a certain extent during postprocessing by following the B_1 map up its gradient. Keeping track of the flip-angle dependence through its cyclic variations up the B_1 gradient allows interpolation of the points near θ_1 flip angles of 90° , 270° , etc., and correct, single-valued determination for the θ_1 angles greater than 180° . Using this method, the B_1 map could be generated at points closer to the wires of the surface coil. This method fails, however, when the variations in the

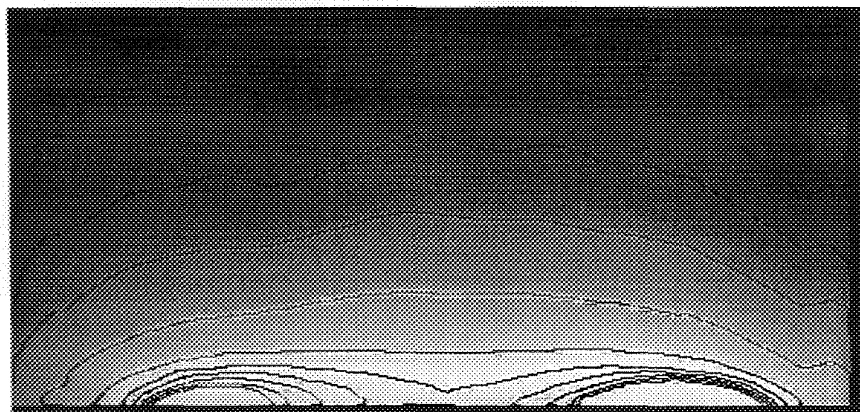


FIG. 4. The B_1 map of an 8 cm diameter surface coil made using a 1 cm thick rectangular phantom filled with 10 mM copper sulfate in water. The images used to construct the map were made at an echo time of 22 ms and a repetition time of 1 s. The phantom was aligned perpendicular to both the main magnetic field and the axis of the surface coil. The lowest contour is at a field value of 0.5 G. The contours are separated by 0.3 G. The field was not determined inside the 3.5 G contour because the variation in the flip angle was beyond the resolution of the images.

flip angle of the map are spatially smaller than the resolution defined by the pixel size in the 256×128 images. This was also a problem, although much less apparent, with the volume-coil map. Nevertheless, an accurate B_1 map could be easily generated without any postprocessing in a 22-strut discrete cosine coil of 13.5 cm diameter to within 3 mm of the wires with the greatest current and even closer to the other wires. In the surface-coil B_1 map, the B_1 values could be accurately determined to within 8 mm of the wires in an 8 cm diameter coil without postprocessing and to a range of 5 mm with postprocessing. Generating B_1 maps that are accurate to a closer limit is simply a matter of making a set of higher resolution images. In all maps presented, the data were subjected to a median filter with a 5×5 kernel in order to keep the contour levels from varying with the noise.

There have been several other methods of constructing B_1 maps published in the past several years (6–8). The method of Stollberger *et al.* (6) uses the ratio of a $\theta_1 - \tau - 180^\circ - \tau$ spin-echo image and a $\theta_2 - \tau - 180^\circ - \tau$ spin-echo image to construct the map. This method has some similarities to the volume-coil method presented here; however, the earlier method is less sensitive to the small changes in B_1 fields in volume coils because the signal generated follows a less steep $\sin \theta$ function. The method of Murphy-Boesch *et al.* (7) uses nutation rates to calculate B_1 maps. This method, despite being very precise, is applicable only on phantoms of uniform proton density and has been implemented only to measure B_1 fields in a single dimension. The method presented here is fully two dimensional and is applicable to samples of non-uniform proton density. The method of Oh *et al.* (8) relies on a $90^\circ_x - 180^\circ_x - 90^\circ_y$ composite-spin-echo sequence

which creates a phase change that is dependent on the field intensity at that point in the image. This method is not only very elegant, but it has the additional benefit of generating a phase map of the object as one of the preliminary steps to calculating the B_1 map. Therefore, although the method presented is certainly not the only method for constructing B_1 maps, it is certainly a simple and fast means of doing so.

ACKNOWLEDGMENTS

This work was supported by NIH Grant RR02305. Special thanks are extended to Dr. John S. Leigh and Dr. Ravinder Reddy for their helpful comments.

REFERENCES

1. G. Bodenhausen, R. Freeman, and D. L. Turner, *J. Magn. Reson.* **27**, 511 (1977).
2. M. R. Bendell and R. E. Gordon, *J. Magn. Reson.* **53**, 365 (1983).
3. L. Bolinger, M. G. Prammer, and J. S. Leigh, Jr., *J. Magn. Reson.* **81**, 162 (1988).
4. M. D. Schnall, V. H. Subramanian, J. S. Leigh, Jr., L. Guylai, A. McLaughlin, and B. Chance, *J. Magn. Reson.* **63**, 401 (1985).
5. D. A. Roberts, E. K. Insko, and J. S. Leigh, Jr., *J. Magn. Reson. Imaging Suppl.* **2**, S27 (1992).
6. R. Stollberger, P. Wach, G. McKinnon, E. Justich, and F. Ebner, Abstracts of the Society of Magnetic Resonance in Medicine, 7th Annual Meeting, Works in Progress, p. 106, 1988.
7. J. Murphy-Boesch, G. J. So, and T. L. James, *J. Magn. Reson.* **73**, 293 (1987).
8. C. H. Oh, S. K. Hilal, Z. H. Cho, and I. K. Mun, *Magn. Reson. Imaging (UK)* **8**, 21 (1990).